Sina Malakouti

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EDUCATION

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran 2015-Present
B.Sc. in Computer Science (Software Engineering) GPA 17.58/20 – 119/140 credits

FIELDS OF INTEREST

Machine Learning, Deep Learning, Natural Language Processing

TECHNICAL SKILLS

PROGRAMMING LANGUAGE
Java, Python, C/C++

WEB PROGRAMMING
Node.js, PHP, JETTY, JavaScript, jQuery, Vue.js, HTML, CSS

SMARTPHONE APP DEVELOPMENT
ReactNative, Android, Objective C

DATABASE
MySQL, MongoDB, SQLite

MACHINE LEARNING & DATA SCIENCE
MATLAB, R, TensorFlow, Deeplearning4j, WEKA, scikit-learn, Pandas, NumPy

OPERATING SYSTEM
Linux, Mac OS X, Windows

RESEARCH EXPERIENCE

Research Assistant Intern - Johannes Gutenberg University, Mainz, Germany (July - Oct 2018)

- Data Mining lab under the supervision of Prof. Stefan Kramer and Zahra Ahmadi

The aim of this project was designing a new symbolic deep architecture based on Hoeffding trees which we **published in ECML PKDD 2019**.

Research Assistant Intern , Tehran, Iran (June 2019 - Present)

- Center for Biomedical and Health Informatics under the supervision of Dr. Mehrdad Oveisi
- Shahid Rajaei Cardiovascular, Medical & Research Center, Tehran, Iran

Applying Machine Learning to solve biomedicals problems, in particular cardiovascular diseases.

Undergraduate Research Assistant, Tehran, Iran (June 2017 – Present)

- Big Data lab under the supervision of Dr. Maryam Amir Haeri
- Amirkabir University of Technology, Tehran, Iran

Working on Ensemble and merging of Decision Trees based on data distribution and Density Estimation in order to achieve a single model.

WORK EXPERIENCE

Software Engineer, Papillon (June – December 2016)
Papillon is a recipe social media for Iranian users with more than 1,000,000 users. My team and I worked on developing this application for Apple product users.

PUBLICATIONS

Ahamdi, Sina Malakouti, Kramer. "DeepTreeNetworks: A New Symbolic Deep Architecture." *Deep Continuous-Discrete Machine Learning (DeCoDeML)* ***ECML/PKDD 2019*** *workshop* (paper)

- I presented this paper at ECML/PKDD 2019.
- This project is still in process, and full paper will be submitted soon.

SELECTED OPEN SOURCE AND COURSE PROJECTS

Human face recognition using Eigen-face method

Classica; & Modern, Local Search Package - GITHUB

Inverted Index Search, tree structures, and hash table data structure library - GITHUB

Full stack online food ordering website - GITHUB

Multi-threading with AI 2D graphical strategic game - GITHUB

TEACHING EXPERIENCE

TEACHING ASSISTANT

Foundation of Data Mining, Linear Algebra, Probabilities & Statistics, Design of Algorithms, Data Structures & Algorithms, Principles of Database Design, Web Programming, Advance Programming (Java)

RELATED COURSES

University Courses

Foundation of Data Mining, Probabilities & Statistics, Linear Algebra, Signals & Systems, Stochastic Processes, Artificial Intelligence and Expert Systems, Design of Algorithms, Data Struct. & Algorithms, Principles of Database Design, and Discrete Mathematics.

Online Courses

Machine Learning (Andrew Ng, Stanford), **Natural Language Processing** (Higher School of Economics), **Algorithm Specialization** (Tim Roughgarden, Stanford), **Mathematics for Machine Learning: Linear Algebra** (Imperial College London), **Basic Statistic** (Matthijs Rooduijn et al. University of Amsterdam)

LEADERSHIP ACTIVITIES

Elected as the President of Student Scientific Chapter of Computer Engineering and IT department, Amirkabir University of Technology – (January 2017- March 2018)

My team and I held more than 70 national and international events, contests, seminars, and symposiums.

- Executive Director of the 16th Amirkabir International Collegiate Programming Contest (ACM ICPC) jointly with KTH – Royal Institute of Technology & TUM- Technical University of Munich.
- Executive Director committee of the 1st Amirkabir National Data Mining Contest.

HONORS & AWARDS

- Selected as an Outstanding Student (GPA based) and eligible to choose Second Major, Amirkabir University of Technology (2016 - Present) - certificate available upon request.
- Ranked top 0.8% of Iranian University Entrance Exam among 167868 students.

Languages

English (TOEFL iBT Score: 106/ 120), **Farsi** (Native)

REFERENCES

- **Dr. Maryam Amir Haeri**, Amirkabir University of Technology: haeri@aut.ac.ir
- **Prof. Stefan Kramer**, Johannes Gutenberg University: kramer@informatik.uni-mainz.de
- **Dr. Mehrdad Oveisi**, Center of Biomedical and Health Informatics and Sessional Lecturer at University of British Columbia: movieisi@cs.ubc.ca
- **Zahra Ahmadi**, Johannes Gutenberg University: zaahmadi@informatik.uni-mainz.de